

JAY CONSEJO

Full-Stack Builder & AI-First Founder

Project Portfolio | May 2026

Six products shipped end-to-end across AI, Web3, and full-stack web — from a self-hosted AI CTO with voice-first cockpit and a 5-entity agent constellation, to an on-chain campaign verification platform on Base, to an AI video production studio automating YouTube workflows. Each system was architected, built, deployed, and operated as a solo founder using Claude Code as the daily build partner. Production stacks ship Solidity contracts on Base mainnet, tRPC-typed Next.js 16 apps, FastAPI services with Supabase RLS, and a self-hosted multi-agent orchestration substrate.

ALEX CTO

AI CTO & AGENT CONSTELLATION

Self-hosted AI CTO running as a 5-entity agent constellation. ALEX joins meetings, speaks in real-time over WebRTC, manages calendar + Slack + email, RAGs project state across 7 active projects, and coordinates with ATHENA (an 18-agent development agency) via a versioned correspondence vault. CEO inbox at /ceo surfaces decisions + asks + plain-English digests.

BRAIN	Python 3.11 + FastAPI, Claude Opus 4.7 (Max OAuth subprocess rail)	COCKPIT	Next.js 16 (App Router, Turbopack), React 19, TS 5
VOICE	Pipecat + Gemini Live API, Cloudflare Realtime SFU (WebRTC)	RAG	ChromaDB + ONNX embeddings (sentence-transformers retired)
ROUTING	5-tier multi-model: Ollama → Haiku → Sonnet → Opus / Codex GPT-5.5	LEAD-GEN VOICE	ElevenLabs Conversational AI (Sofia agent, dual persona)
DATABASE	PostgreSQL (telemetry + decision log JSONL)	COORDINATION	constellation-vault git submodule (cross-Claude turns)
HOSTING	GCP e2-small + Docker compose + Tailscale SSH	EDGE	Cloudflare Tunnel + Access (JWT-gated, service-token RBAC)

CONSTELLATION ARCHITECTURE (5 ENTITIES)

- **Jay (CEO) + Jay-twin** (digital second brain at /ceo) — decision substrate & CEO oversight surface
- **ALEX** (CTO/brain) — orchestrator, voice peer, project state RAG, calendar/email/Slack interface
- **Sofia** (twin personas) — lead-gen voice on ElevenLabs + EA voice on Pipecat, sharing one prompt vault
- **ATHENA** (18-agent dev agency) — autonomous pipeline: planner → coder cascade → cross-model reviewer → PR

KEY ARCHITECTURE DECISIONS

- Subscription rail (Max OAuth) for personal-use orchestration — \$220/mo flat across two systems, \$0/call
- Cross-model reviewer rule — reviewer model must differ from coder model (eliminates same-family blind spots)
- Decompose-before-fire — sprint envelopes >~400 LOC OR >2 dirs at issue-write trigger mandatory pre-decomposition
- Bilateral coordination via versioned vault — every cross-system turn is a git commit, atomic and auditable
- 67+ ADRs locked across the system — every non-trivial decision is referenced, not absorbed silently

Murmura Studio

AI VIDEO PRODUCTION

AI-powered video production platform for YouTube creators. Automates the full content pipeline — research, scripting, voiceover, rendering, and publishing.

FRAMEWORK	Next.js 16 (App Router, Turbopack)	FRONTEND	React 19, TypeScript 5, Tailwind CSS 4
AI / LLM	Vercel AI SDK 6, Anthropic, Google Gemini	VIDEO	Remotion 4 (programmatic rendering)
VOICE	ElevenLabs, Edge TTS, Google Cloud TTS	AUDIO	Deepgram (speech processing)
DATABASE	PostgreSQL (Railway) + Prisma 7 ORM	STORAGE	Cloudflare R2 (S3-compatible)
AUTH	NextAuth.js 5 (Google OAuth)	DEPLOY	Vercel (serverless, 5-300s functions)

KEY ARCHITECTURE DECISIONS

- Server Components for data-heavy pages, client components only where interactivity requires it
- Vercel AI SDK streamText + tool-calling for autonomous content workflows
- Cloudflare R2 over S3 — zero egress costs for media-heavy platform
- Cost-optimized AI cascade: free tiers first, paid only on overflow

On-chain campaign verification platform on Base. Brands create campaigns, creators submit proof of social content, and rewards are distributed via smart contracts with cryptographic verification.

FRAMEWORK	Next.js 14 (App Router)	FRONTEND	React 18, TypeScript, Tailwind CSS 3
API LAYER	tRPC 11 (end-to-end type-safe RPCs)	WEB3	Wagmi 2, Viem 2, RainbowKit 2
AUTH	SIWE (Sign-In with Ethereum)	CONTRACTS	Solidity 0.8.27, Hardhat, OpenZeppelin v5
CHAIN	Base (mainnet + Sepolia testnet)	DATABASE	PostgreSQL (Railway) + Prisma 7 ORM
CACHE	Upstash Redis (rate limiting)	DEPLOY	Vercel + Base (on-chain)

SMART CONTRACTS

- **CampaignEscrow** — ERC20 deposits, ECDSA-verified claims, reentrancy protection
- **CampaignBadge** — ERC1155 achievement badges with IPFS metadata
- Deployed on Base mainnet and Sepolia testnet via Hardhat

KEY ARCHITECTURE DECISIONS

- tRPC for fully type-safe API — no REST schema drift between frontend and backend
- SIWE over traditional auth — wallet-native identity, no password management
- ECDSA signature verification on-chain — server signs, contract verifies (gas-efficient)
- Base chain for low fees and Coinbase ecosystem alignment

Decoded Daily News

AI NEWS PLATFORM

AI-personalized daily news digest for professionals. Fetches from 40+ sources, uses multi-step AI summarization, and delivers role-personalized digests via email and web at 6 AM daily.

DAILY DECODED

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BACKEND	Python FastAPI (async-first)	FRONTEND	HTML/CSS/JS, Tailwind CSS (CDN)
AI / LLM	Gemini, Vertex AI, Groq, Claude Haiku	DATABASE	Supabase (PostgreSQL + Auth + RLS)
SCHEDULER	APScheduler (in-process cron)	EMAIL	SendGrid (HTML + PDF attachments)
SCRAPING	RSS + NewsData.io + Trafilatura	PDF	ReportLab (branded digest PDFs)
BACKEND HOST	Railway PaaS (staging + production)	FRONTEND HOST	Cloudflare Pages (global CDN)

- KEY ARCHITECTURE DECISIONS
- Supabase Row-Level Security — frontend uses anon key safely, all access controlled at DB level
 - Hash-pinned dependencies (SHA-256) after supply chain incident with litellm package
 - 2-step legal-safe AI summarization — max 3 consecutive source words, original analysis only
 - AI provider cascade: Gemini free → Vertex AI → Groq → Claude Haiku (cost optimization)
 - Cloudflare Pages for frontend — zero cost, global CDN, automatic HTTPS

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PORTFOLIO & AGENCY SITE

Personal portfolio and AI agency landing page. First project built entirely through the HADP agency pipeline from design through deployment.

FRAMEWORK	Next.js 16 (App Router, Turbopack)	FRONTEND	React 19, TypeScript 5, Tailwind CSS 4
UI	shadcn/ui + Base UI React, Lucide icons	DEPLOY	Vercel (automatic Git deploys)

- KEY ARCHITECTURE DECISIONS
- Design-neutral skeleton — structural foundation reskinable via CSS variables
 - Full SEO infrastructure: Metadata API (OG/Twitter), sitemap.ts, robots.ts
 - 6 security headers: HSTS, X-Frame-Options, nosniff, referrer, permissions policy

ATHENA (formerly HADP)

AUTONOMOUS DEV AGENCY

Self-hosted multi-agent software development agency — 18 specialized agents across 3 teams (Engineering, Marketing, Creative) running an autonomous planner → coder → reviewer pipeline. Operates as a peer node in the ALEX constellation: receives sprint tickets, fires the cascade, opens PRs, and reports back via a versioned correspondence vault. First production sprint shipped in 7 sessions; cascades land within tier=standard 600s budget.

ORCHESTRATOR	Codex GPT-5.5 (Codex Pro sub OAuth) on Windows	CODER CASCADE	DeepSeek-Flash → DeepSeek-Pro (NIM) → Codex fallback
REVIEWER	Cross-model: Opus 4.7 (when coder=Codex), Codex (when coder=Opus)	LOCAL MODELS	Llama — Qwen 2.5 14B classifier, phi4-mini Tier-S exec
COORDINATION	Constellation-vault submodule (turns + ADRs + glossary)	COMMS	Slack (bot + webhooks), CEO /ceo inbox at alex.jayconsejo.com
NETWORKING	Cloudflare Tunnel + Access, Tailscale	HARDWARE	PC Ryzen 7 5700G + RTX 5070 (Windows 11)

- KEY ARCHITECTURE DECISIONS
- Constellation pivot — HADP's 19-agent orchestration evolved into ATHENA as a peer node of ALEX (CTO/brain), not a hierarchy
 - Decompose-before-fire rule — pre-fire intake gate refuses oversized envelopes; first measurement shipped clean at ~489s under 600s cap
 - Last-manual-finish-at-n=2 — bright line drawn at Sprint 2a; future cascade exhausts trigger Path A re-decomposition, never operator stash-finish
 - Cross-system bilateral substrate — every coordination turn lands as a vault commit; drafts in Research/active/ are not signal until pushed
 - Subscription rails only — Codex Pro + Max + GCP credits cover the entire stack; zero API spend on personal-use orchestration